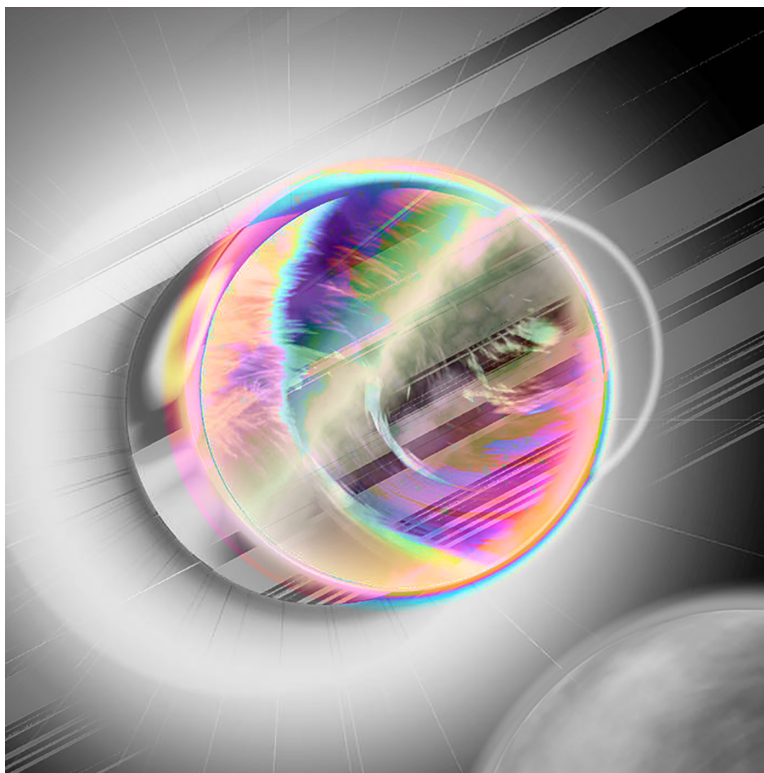


MPLANETARIAN

STREAM OF FREQUENCY WORKSTATION SUITE



Mix Archive Manager

v0.3.0 Standard Reference Manual

Cross-Platform Media Orchestration Console & Lossless FLAC Mastering Suite

Platforms: Linux (Bazzite / SteamOS / Fedora) • macOS (Sequoia / Sonoma) • Windows 10/11 • FreeBSD

Core Architecture: Bash 5.x • Python 3.11+ • 32-Bit Lossless FLAC • MIT License

System Architecture & Highlights

Mix Archive Manager (MP_Mix_Manager_v0.3) is a professional workstation orchestration console designed for high-resolution audio mastering, multi-hour DJ recording curation, Traktor Pro library management, live stream monitoring, and local AI workflow automation across Linux, macOS, Windows, and FreeBSD.

SPOTLIGHT: MPlanetaryrians Alarm Clock (Wake Up Edition)

- **Lossless FLAC Wake-Up:** Selects a random mix from local archive drives and executes a 3-minute linear volume ramp (0% → 100%) in Strawberry or cliamp.
- **Active Motion Verification:** Detects physical mouse movements to silence the alarm. If stationary for 10 minutes, an alternate mix is triggered.
- **Right-Monitor Browser Geometry:** Calculates active display geometries and opens your morning browser window on the display to the right.
- **Steam Breakfast Selector:** Scans local libraryfolders.vdf files and presents 3 random video games to launch directly via one keypress.

SPOTLIGHT: Traktor Pro Live Session History to 4-Deck Prep

- **Automated Discovery:** Automatically locates and reads collection.nml and History.nml across Traktor 3 and Traktor Pro 4 without manual path setup.
- **Harmonic Key Sorting:** Reconstructs past live setlists in ascending Camelot musical key order (1A–12B) to guarantee compatible initial transitions.
- **Root Collection Injection:** Generates a verified NML playlist directly into Traktor's root library tree (\$ROOT).
- **Full-Screen Stage Launch:** Brings Traktor Pro to the foreground in full-screen mode and pre-loads Decks A, B, C, and D with the first four tracks ready to mix.

Core Feature Matrix (78 Console Operations)

Section 1: Mix Archive Workflow & Ingestion

#	Operation	Description
01	Run FLAC Conversion Process	Concatenates split WAV files, master encodes to 32-bit FLAC (-sample_fmt s32 -compression_level 12), embeds resized artwork, and outputs acoustic spectrograms.
02	Convert Formats & Split FLACs	Transcodes audio between MP3 (320k, V0, 256k), Ogg, Opus, AAC, ALAC, and WAV bit depths; includes sample-accurate lossless FLAC splitting into equal segments.
03	Retrieve Unconverted WAVs	Scans archive directories and isolates WAV stems missing a processed master FLAC counterpart.
04	Search & Import Mixes	Auto-discovers external USB drives, local storage, and SMB network shares; filters files >= 100MB, deduplicates, and transfers with live logs.
05	Rename Mix Assets	Atomically renames FLAC audio, TXT tracklists, and spectrogram PNGs uniformly across all archive directories.
06	Duplicate File Removal	Executes fast chunk-based hashing to identify identical mixes and moves them to quarantine or deletes them.
07	Export/Package Mixes	Copies complete release packages (FLAC + Cover Art + TXT/HTML/PDF Tracklists + Spectrograms) to external storage drives.
08	Manage SHA-256 Checksums	Generates and validates checksums.sha256 manifests across audio libraries to protect archives against bit rot.
09	Verify FLAC Integrity	Runs multi-threaded decode passes across all CPU cores to detect and quarantine damaged FLAC bitstreams.
10	Cloud & Remote Backup	Automated backup suite targeting Google Drive, iCloud, Dropbox, and external directories with companion asset support.
11	Mix Storage Drive Space	Displays filesystem mount capacity, active usage progress bar, and hosted audio file counts on primary archive volumes.
12	Refresh Archive Status	Clears cache and performs full recounts of pending WAVs, converted WAVs, FLAC masters, and tracklists.
13	Configure Storage Path	Sets default storage path override (defaults to /run/media/\$USER/WD_BLACK_B/MIX_ARCHIVE/ or MIX_ARCHIVE/).

Section 2: Tracklist, Metadata & Promotion

#	Operation	Description
14	Tracklist Suite	Browses, searches, and exports tracklists to styled HTML and printable vector PDF documents.
15	Mix Search & Live Playback	Searches archive catalog, initiates cliamp playback, and displays live setlists in the console.
16	Scan Missing Tracklists	Parses Traktor Pro collection.nml history files to generate timestamped text tracklists.
17	Master Tracklist Index	Compiles all individual tracklists into a single, unified searchable HTML index file.
18	MusicBrainz Picard	Launches Picard audio tagger; automatically installs missing dependencies across platforms.
19	Promo & Outreach Emailer	Formats and delivers promotional emails to labels, publishers, and club promoters with artwork and cue attachments.
20	Multi-Platform Syndication	Release scheduler managing Apple Podcasts, Spotify, YouTube, SoundCloud, Mixcloud, DI.FM, Proton Radio, and Bandcamp with .ics / RSS exports.
21	Music Store Quick-Launch	Parallel browser dispatcher for Beatport, Apple Music, and Bandcamp store portals.

Section 3: Audio Playback, DAWs & Sound Suite

#	Operation	Description
22	DAWs Launch Hub	Unified launcher for REAPER, Logic Pro, FL Studio, Traktor Pro, GarageBand, Ardour, LMMS, Bitwig, and Bespoke Synth.
23	Open Mix in DAW	Searches mixes by keyword and opens stems directly in REAPER, Logic Pro, FL Studio, or Audacity.
24	Spectrogram Analysis Suite	Renders in-place acoustic spectrograms using Spek, SoX 24-bit colormaps, Sonic Visualiser, and Praat.
25	Audacity Audio Editor	Launches Audacity or installs via Homebrew, Flatpak, winget, or native package managers.
26	Audio Players Menu	Quick launch hub for cliamp, Strawberry, VLC, foobar2000, Winamp, Apple Music, Haruna, and Kodi.
27	Configure Audio Engine	Sets default audio/video players, startup auto-play toggles, and console tracklist preferences.
28	cliamp Terminal Player	Built-in retro CLI audio player featuring cross-platform clipboard copying (pbcopy, clip.exe, wl-copy).
29	Audio Stream Specs HUD	Probes bit depth (16/24/32-bit), sample rate (48 kHz), bitrate, compression ratios, and paired companion assets.
30	Archive Playlists Suite	Builds, synchronizes, and exports .m3u8 and .xspf playlists to Strawberry, VLC, mpv, or Kodi.
31	Strawberry Music Player	Opens Strawberry Music Player in a separate desktop window.
32	VLC Media Player	Launches VLC audio/video media player.
33	Traktor Harmonic Prep / Haruna	macOS: Sorts Traktor session history into harmonic Camelot keys (1A-12B), injects into root, and pre-loads Decks A-D. Linux: Launches Haruna.
34	Kodi Media Center	Launches Kodi home theater software.
35	Connected USB MIDI Devices	Enumerates connected MIDI synthesizers, DJ controllers, and hardware control surfaces.
36	Studio Hardware Diagnostics	Probes active PipeWire/ALSA audio sinks, sample rates, buffer latencies, and connected studio interfaces.
37	Master Audio Volume Toggle	Instantly mutes or unmutes system master audio via PipeWire wpctl, pactl, or AppleScript.
38	Morning Alarm Clock	Morning wake-up alarm with lossless volume ramp, mouse motion wake detection, and Steam breakfast game launcher.

Section 4: Video Production, Art & Visual Media

#	Operation	Description
39	Generate YouTube Video	Encodes 4K UHD, 1080p, and 720p YouTube videos using NVENC/VideoToolbox hardware acceleration and 320k AAC audio.
40	Cut or Split Video File	Precision timestamp cutter and equal-segment video splitter supporting stream-copy or re-encoding.
41	Launch Video Playlists	Launches Defasten or visual companion playlists in VLC, mpv, or Haruna.
42	Launch Specific Video	Dispatches user-specified video files or streaming URLs to the default video playback engine.
43	GIMP Image Editor	Launches GIMP image editor or installs missing dependencies via system package manager.
44	Convert Cover Art	Converts artwork between JPEG, WebP, PNG, and TIFF with strict binary byte budgeting (<= 1MB podcast standard).
45	View Cover by Mix Number	Searches COVERS/ directory by episode number and opens the artwork in the system image viewer.
46	Procedural Netpbm Art	Pure algorithmic P6 binary PPM art engine: renders mathematical gradients (linear, plasma, angular) with typography overlays.
47	Electric Sheep Screensaver	Launches Electric Sheep generative screensaver visualizer.
48	Sync Video Companion	Background daemon monitoring audio playback to auto-launch and sync matching video companions.
49	GPU Screen Recorder (Linux)	Launches hardware-accelerated DJ set video capture via gpu-screen-recorder-gtk or Flatpak.

Section 5: Live Monitors & System Diagnostics

#	Operation	Description
50	Live Tracklist Monitor	Real-time console display tracking Strawberry MPRIS and cliamp playback progress.
51	Traktor Live Monitor	Cross-platform monitor displaying Traktor CPU %, RAM, deck states, active recording file locks, and AppleScript triggers.
52	Live File Transfer Monitor	Inspects transfer speeds, byte positions, and percentage for huge multi-gigabyte files.
53	Chrome Upload Monitor	Monitors web browser upload processes (Apple Podcasts Connect, YouTube Studio) in real time.
54	Advanced Archive Statistics	Calculates total audio duration, file counts, storage footprint, and tracklist completeness.
55	View Running Background Tasks	Scans process tables for active ffmpeg transcode passes, cloud syncs, or AI jobs.
56	System Resource Monitor	Launches btop for multi-core CPU and memory profiling.
57	GPU Process Monitor	Launches nvidia-smi for real-time monitoring of NVIDIA GPU clock, VRAM, and power draw.
58	Standard Process Viewer	Quick-launches top inside the active manager terminal session.

Section 6: System, Network & Remote Hardware

#	Operation	Description
59	WAN2GP Video AI Server	Manages WAN2GP AI video server profiles (Flux Klein 9B, LTX Video 2B/13B).
60	Beszel Monitoring Suite	Manages Beszel server monitoring hub and hardware monitoring agent daemons.
61	Ollama LLM Container Server	Controls Ollama container instances (Qwen 2.5, Llama 3.1) with interactive chat options.
62	DeepSeek Harness (dsh-mobile)	Controls DeepSeek mobile web portal (pnpm dsh web) over local LAN (port 3080).
63	Network & Congen KDE Connect	Controls SSH, Samba, and FTP daemons, plus the Congen KDE Connect smartphone remote controller.
64	Block Internet Access	Enables isolated firewall rules blocking WAN traffic while preserving local LAN transfers.
65	Restore Internet Access	Restores immediate full internet connectivity.
66	Display Settings (OS Tailored)	Opens Plasma Wayland display settings, macOS Display Settings, or Windows ms-settings:display.
67	Audio Settings (OS Tailored)	Opens Plasma X11, Audio MIDI Setup on macOS, or Windows Sound Panel.
68	Close Desktop Applications	Gracefully closes external desktop windows using AppleScript, PowerShell, or wmcctl.
69	System Maintenance & Cleanup	Executes platform maintenance (Linux ujust clean-system, macOS brew cleanup, Windows TRIM/winget, FreeBSD pkg clean).
70	GeeXLab Demo Launcher	Runs 3D/OpenGL shader demos and GPU stress benchmarks.
71	Burn ISO to USB Drive	Writes bootable ISO images directly to removable USB storage with verification passes.
72	Dynamic System MOTD Banner	Generates stylized ANSI MOTD table summarizing recent mixes, sizes, formats, and hardware vitals.

Section 7: AI Engine, Shell CLI & Settings

#	Operation	Description
73	AI Assistant CLI (AGY)	Antigravity CLI assistant interface managing Ollama, DeepSeek, and remote models.
74	Interactive Bash Subshell	Built-in interactive Bash shell and direct command execution runner.
75	Terminal Theme Switcher	Switch between 9 retro terminal palettes (Dracula, Nord, Matrix, Cyberpunk, Solarized, Tokyo Night, Monokai, Gruvbox, Emerald).
76	Config Migration & Snapshots	Configuration backup and restore engine; exports portable .tar.gz bundles and handles automated updater checks.
77	Reboot Workstation	Cross-platform system reboot with safety confirmation dialogs.
78	Exit Manager	Cleans up and exits the console session.